Lecture 10.

→ summary of the Partial Collapse Story

→ Begin to explore properties & uses of entangled states.

→ Bell Basis for two-qubit states.

(all entangled states)

Given a state $|\psi\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$

↳ we want to measure q_1, rewrite in this way:

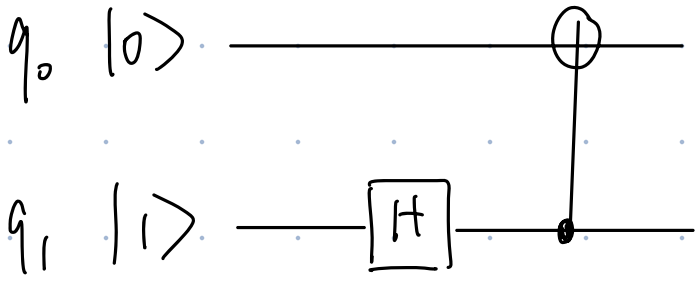
$$|\psi\rangle = \sqrt{p_+}|0\rangle \otimes \left[\frac{\psi_{00}|0\rangle + \psi_{01}|1\rangle}{\sqrt{p_+}}\right] +$$

$$\sqrt{p_-}|1\rangle \otimes \left[\frac{\psi_{10}|0\rangle + \psi_{11}|1\rangle}{\sqrt{p_-}}\right]$$

normalize such that both are unit vectors.

- Measurement reduce entanglement.

→



$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

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↳ It's impossible to produce entanglement from local operations & classical communication ~~from~~ on product states

Bell basis for 2 qubits

$$|B_0\rangle = \frac{1}{\sqrt{2}}\left[|01\rangle - |10\rangle\right]$$

$$|B_1\rangle = \frac{1}{\sqrt{2}}\left[|01\rangle + |10\rangle\right]$$

$$|B_2\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle - |11\rangle\right]$$

$$|B_3\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle + |11\rangle\right]$$

$$|01\rangle = \frac{1}{\sqrt{2}}\left[|B_0\rangle + |B_1\rangle\right]$$

↳ product state are superposition of entangled states

Spooky Action @ a distance

Consider a product state

$$|\psi\rangle = \underset{\text{Bob}}{|\chi\rangle} \otimes \underset{\text{Alice}}{|0\rangle}$$

Q: How many distinct (i.e. mutually orthogonal) 2-qubit states can Alice create w/ local operations.

Answer: 1 more (2 total)

Alice can perform $\mathbb{X} = I \otimes X$

$$I \otimes X|\psi\rangle = |x\rangle \otimes |1\rangle$$

$|x\rangle \otimes |0\rangle$

$|x\rangle \otimes |1\rangle$

are these 2 orthogonal?

① $|x\rangle \otimes |0\rangle = \begin{pmatrix} a \\ 0 \\ b \\ 0 \end{pmatrix}$

$\begin{pmatrix} a \\ b \end{pmatrix}$

① · ② = 0

② $|x\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ a \\ 0 \\ b \end{pmatrix}$

$\begin{pmatrix} a \\ b \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

<u>Yes!</u>

$$(I \otimes I)|B_0\rangle = |B_0\rangle = \frac{1}{\sqrt{2}}[|01\rangle - |10\rangle]$$

$$(I \otimes Z)|B_0\rangle = \frac{1}{\sqrt{2}}[|01\rangle + |10\rangle]$$

Remember this!

$$= |B_1\rangle$$

$$\begin{cases} Y|0\rangle = i|1\rangle \\ Y|1\rangle = -i|0\rangle \end{cases}$$

$$(I \otimes X)|B_0\rangle = \frac{1}{\sqrt{2}}[|00\rangle - |11\rangle] = |B_2\rangle$$

$$(I \otimes Y)|B_0\rangle = \frac{-i}{\sqrt{2}}[|00\rangle + |11\rangle] = -i|B_3\rangle.$$

so Alice can make 4 orthogonal states

2 independent system → 1 system.

Individual

"qubits don't have their 'own' state"

in an entangled state

for one qubit state $|\psi\rangle = |+\hat{n}\rangle$

need better explanation.

$\langle x\rangle$: x coordinate

$\langle y\rangle$: y coordinate

$\langle z\rangle$: z coordinate

$\langle\psi|I|\psi\rangle = 1$

$\langle\psi|X|\psi\rangle = \hat{n}_x$

$\langle\psi|Y|\psi\rangle = \hat{n}_y$

$\langle\psi|Z|\psi\rangle = \hat{n}_z$

$\langle B_0|I|B_0\rangle = 1$

$\langle B_0|I\otimes X|B_0\rangle = \langle B_0|B_2\rangle = 0$

$\langle B_0|I\otimes Y|B_0\rangle \overset{(-i)}{=} \langle B_0|B_3\rangle = 0$

$\langle B_0|I\otimes Z|B_0\rangle = \langle B_0|B_1\rangle = 0$

there's a 2-sphere representation of 2 qubit states.

it doesn't make sense to talk about the state of a single qubit in an entangled system

Consider: the 2 qubits are perfectly anti-correlated.

$\langle B_0|Z\otimes Z|B_0\rangle = -1$

$$\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}$$

$\langle B_0|X\otimes X|B_0\rangle = -1$

$\langle B_0|Y\otimes Y|B_0\rangle = -1$

no matter what basis they are in, they must be opposites.

parity is odd.

The Killer App?